



BUILDING SIMULATION IN DESIGN PRACTICE

799 Broadway , New York

Location- New York, NY
Instructors- Roya Razaee, Marcelo Bernal, Tyrone Marshall, Perkins & Will
Term- Spring 2020
Team members- Kiran Golla Prerana Kamat
Softwares used- Rhino, Grasshopper, JMP data analysis
Simulation Tool- Butterfly, Honeybee, Ladybug, Octopus, Galapagos

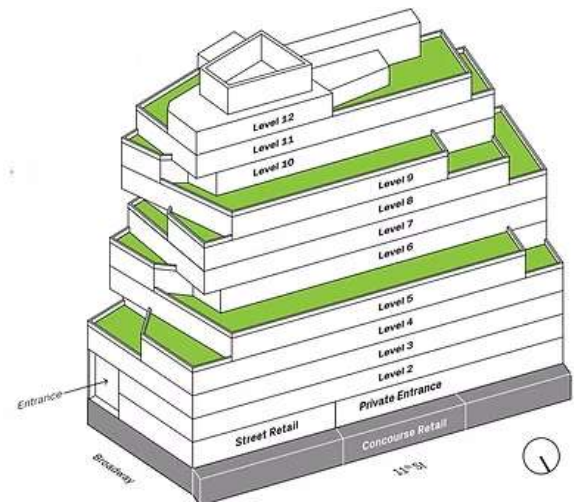
This modeling and simulation course the following domains: Parametric Modeling, Solar, Energy, Air Flow and Ventilation, Daylighting, View, Statistical Analysis, and Data Visualization. computational techniques to generate a large space of design options, simulate a variety of building performances, evaluate and explore the options, and make informed design decisions in a systematic framework called Design Space Construction (DSC) that was developed at Perkins & Will.



The first strand of the course dealt with fundamental knowledge of building modeling and simulation use of parametric tools and introduction to techniques of multi-criteria optimization tested on 799 Broadway Building by Perkins and Will.

PROGRAM AND SPECIFICATION

- 40.732362, -73.991485
- Broadway St, Greenwich Village,
- Manhattan, New York
- The 168,645-square-foot structure
- Retail on the ground and sub-grade levels.
- Offices on upper floors



CEILING	FLOOR	AVAILABILITY	TERRACE
15'	Level 12*	3,642 SF	2,200 SF
15'	Level 11*	7,952 SF	543 SF
15'	Level 10	8,756 SF	2,034 SF
15'	Level 9	11,823 SF	767 SF
15'	Level 8	12,866 SF	1,187 SF
15'	Level 7	14,738 SF	255 SF
15'	Level 6	14,735 SF	3,437 SF
15'	Level 5	19,828 SF	1,114 SF
15'	Level 4	21,409 SF	671 SF
15'	Level 3	22,478 SF	
15'	Level 2	20,821 SF	
17'	Street Retail	19,468 SF	
17'	Concourse Retail	12,670 SF	
TOTAL		182,187 SF	12,208 SF

*20' double-height office space

The second strand comprised of a self-initiated design project focused on the development of an integrated model and the application of a realistic design decision making process.

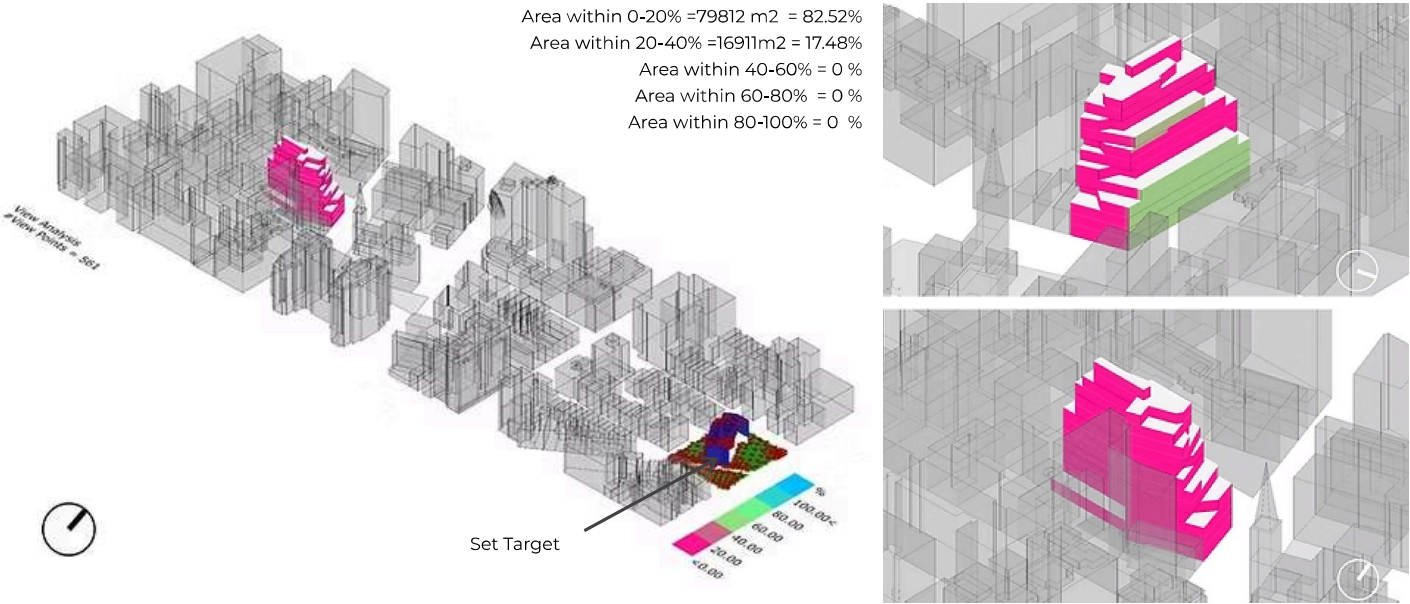
Working in a team, we first formulated the research questions, investigated precedents, and constructed parametric relationships that described the geometry and program configuration using DSC. We then evaluated the design options for multiple objectives, analyzed and visualized the data, extracted the insights, and formulated guidelines for design.

PRECEDENT INVESTIGATION

i. VIEW ANALYSIS

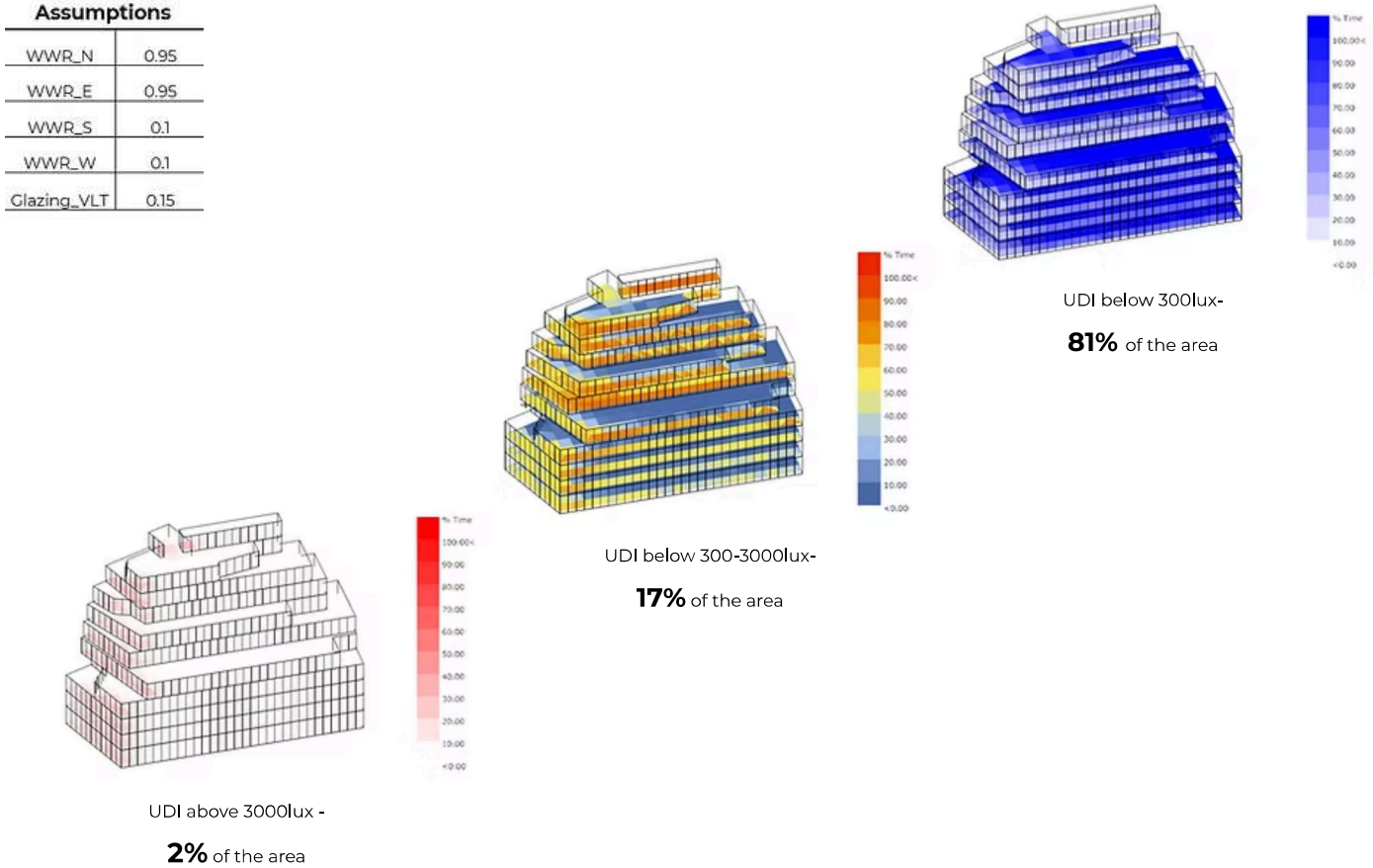
About 82.52% (79812 m2) of the facade lies between a view range of 0-20% to the set target point

The remaining 17.48% lies in the 20-40% view range

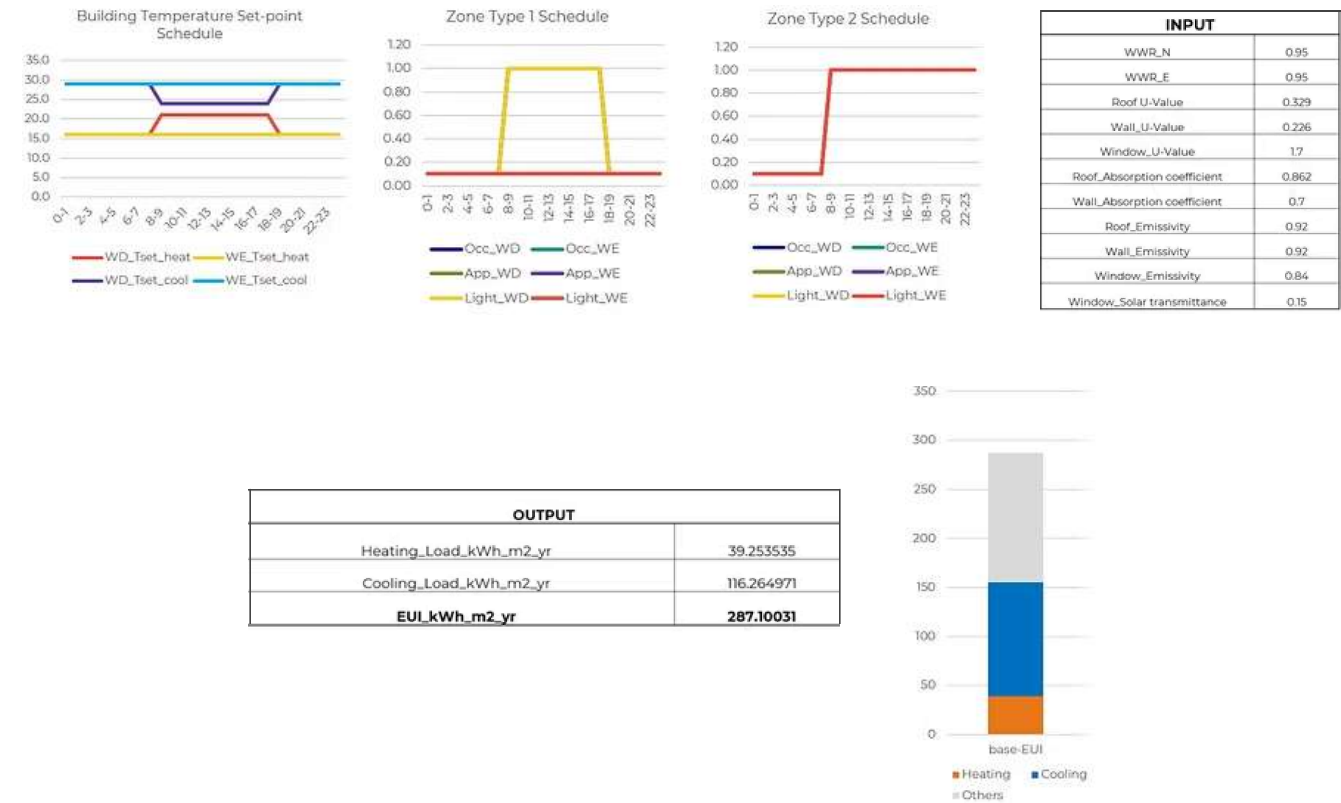


ii. USEFUL DAYLIGHT ILLUMINANCE (UDI)

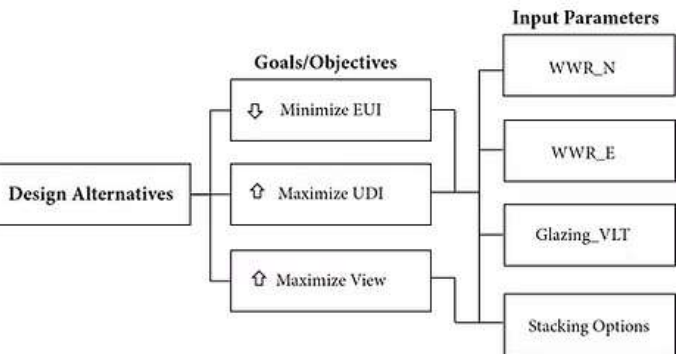
Assumptions	
WWR_N	0.95
WWR_E	0.95
WWR_S	0.1
WWR_W	0.1
Glazing_VLT	0.15



iii. ENERGY USE INTENSITY (EUI)

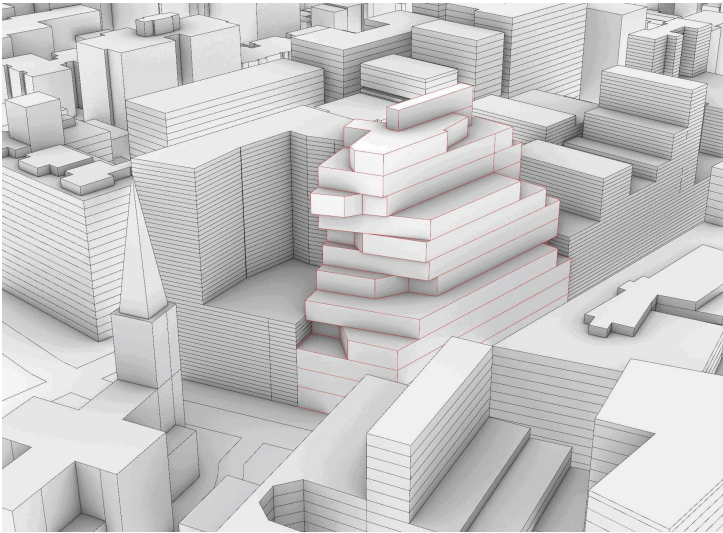
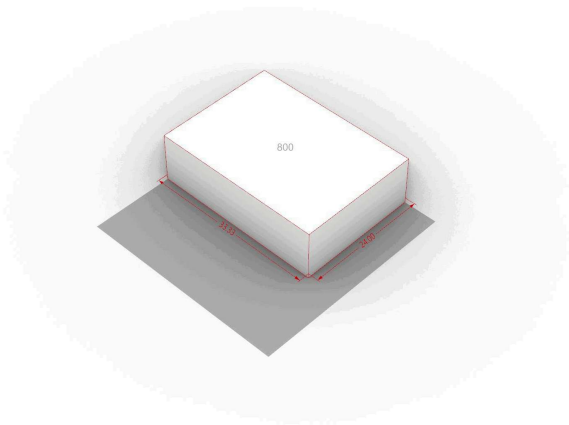


OBJECTIVE FORMULATION



To achieve the objectives of Maximum View and UDI, Minimum EUI, Window to wall Ratio in the North and East facade, Glazing VLT (the amount of visible light waves that transmit through a material) and Stacking options are set as input parameters.

Stacking Options are changed such that the area remains constant while the dimensions of the floor plates are varied. The below animations show the concept thus used on a rectangular plate and how the same is applied on the building.



QUANTIFICATION OF FULL FACTORIAL DESIGN SPACE

Input Parameters	Options	Total
WWR_N	0.95, 0.9, 0.8, 0.7	4
WWR_E	0.95, 0.9, 0.8, 0.7	4
Stacking Option	1, 2, 3, 4	4
Glazing_VLT	0.31, 0.25, 0.18, 0.15	4
Full factorial		256

REDUCTION OF DESIGN SPACE USING DoE

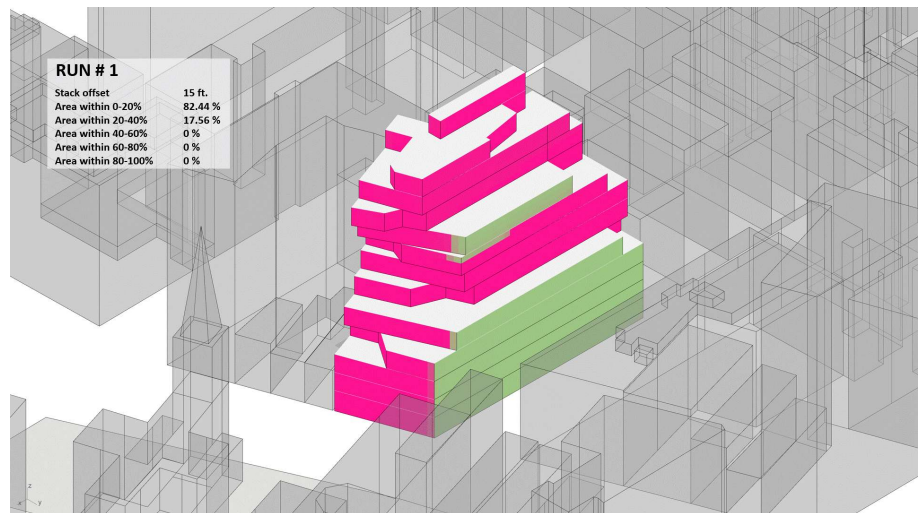
Run	WWR_N	WWR_E	Glazing_VLT	Distance
1	0.95	0.8	0.18	2
2	0.9	0.8	0.31	4
3	0.7	0.7	0.18	1
4	0.7	0.7	0.31	1
5	0.95	0.95	0.15	3
6	0.9	0.7	0.15	2
7	0.8	0.95	0.18	4
8	0.7	0.8	0.25	3
9	0.95	0.9	0.31	1
10	0.9	0.95	0.25	1
11	0.8	0.8	0.15	1
12	0.7	0.95	0.31	2
13	0.95	0.7	0.25	4
14	0.8	0.9	0.25	2
15	0.8	0.7	0.31	3
16	0.7	0.9	0.15	4

The Full factorial of 256 runs is reduced to 16 using DoE which gives a random and varied set of choices using which the analysis is performed

OBJECTIVE 1 : VIEW ANALYSIS

The Set Target point is the same as that of the Precedent's. The analysis is run on the 4 Stacking options which have 15, 18, 21, 24ft offsets.

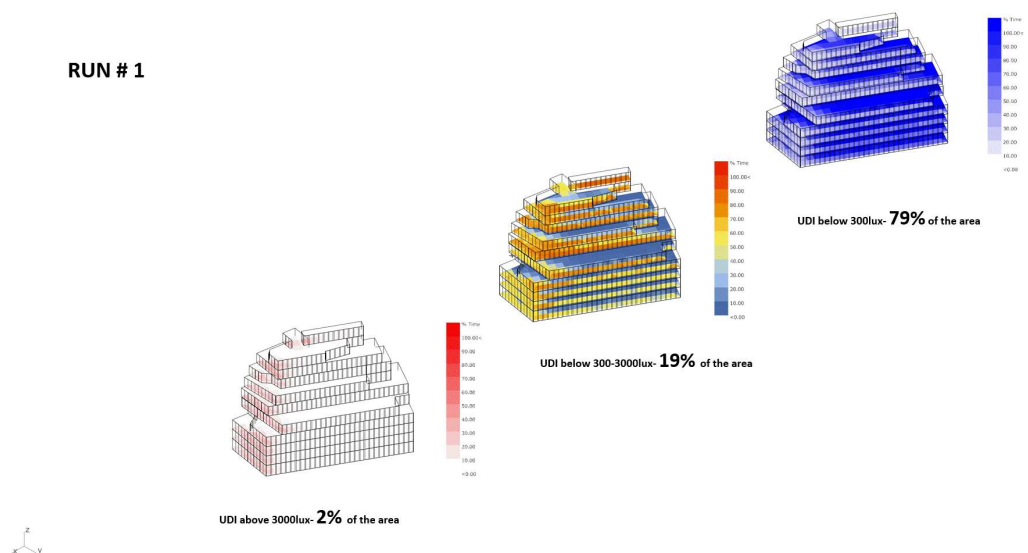
Run 1 with 15ft produces the best result among the options.



OBJECTIVE 2 : UDI

UDI is then run for the 16 different options which have varied input parameters of WWR, Stacking and Glazing VLT.

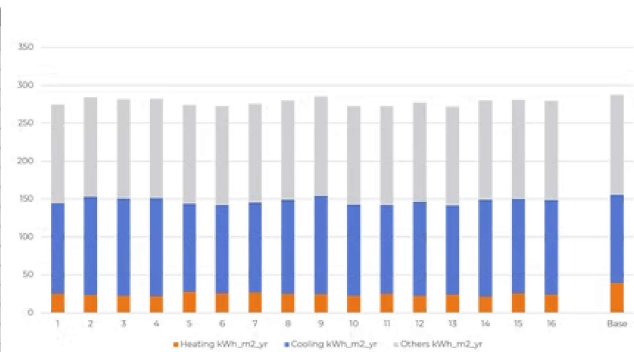
RUN # 1



OBJECTIVE 3 : EUI

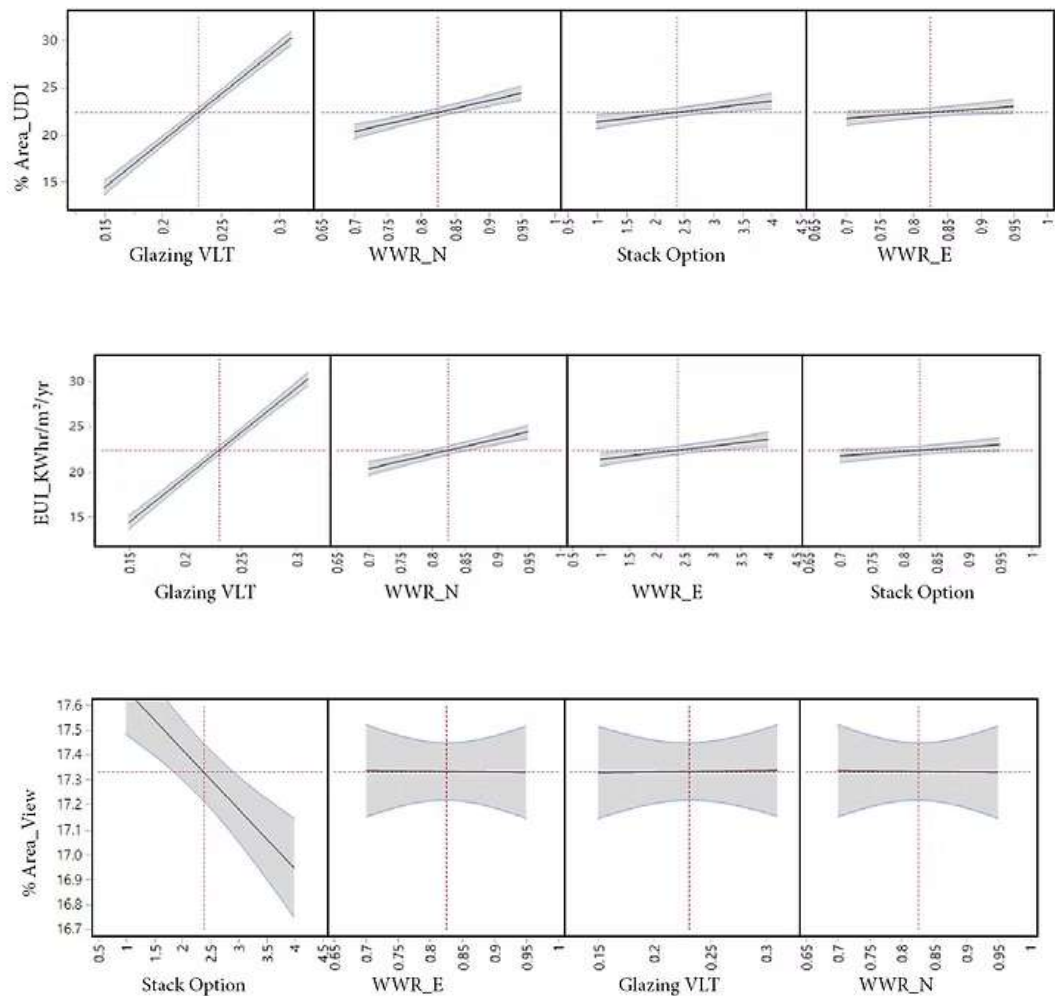
Using the Input Parameters listed below, the Outputs are generated using the grasshopper / honeybee script from which we can notice that the 3th run produces the best output.

Input					Output		
Run#	WWR_N	WWR_E	Glazing_VLT	Distance/ Stacking Option	EUI kWh_m2_yr	Heating kWh_m2_yr	Cooling kWh_m2_yr
1	0.8	0.95	0.18	4	274.564199	25.141916	119.074431
2	0.9	0.8	0.31	4	283.811551	23.773283	128.783626
3	0.8	0.7	0.31	3	281.614059	22.463601	128.108165
4	0.7	0.95	0.31	2	281.972084	22.085929	128.808384
5	0.95	0.95	0.15	3	273.89701	27.726428	115.886674
6	0.9	0.7	0.15	2	272.23382	25.755985	116.358606
7	0.95	0.8	0.18	2	275.385012	26.62764	118.334153
8	0.95	0.7	0.25	4	279.634751	25.061223	123.727492
9	0.95	0.9	0.31	1	285.301114	24.547276	129.354778
10	0.7	0.7	0.18	1	272.356356	22.711728	119.51917
11	0.8	0.8	0.15	1	272.292503	25.339843	116.828218
12	0.7	0.8	0.25	3	277.01534	22.287089	124.137064
13	0.7	0.9	0.15	4	271.626069	24.136686	117.431197
14	0.7	0.7	0.31	1	279.945483	21.17328	127.889082
15	0.9	0.95	0.25	1	280.984621	25.602503	124.404756
16	0.8	0.9	0.25	2	278.951872	23.914461	124.25949
Base	0.95	0.95	0.15	3	287.10031	39.253535	116.264971



SENSITIVITY ANALYSIS

After running the view analysis, UDI and EUI for the 16 iterations/ runs, the model is then taken to PCP / Jmp software to analyse the effect of each of the considered design alternative on the model or understand its sensitivity to the parameters under consideration. The following visuals from the prediction profiler help in understanding the model's sensitivity to each input parameter.



These graphs represent that the UDI and EUI of the building are highly sensitive to the Glazing VLT input while the other inputs like WWR and stack option have little or no effect on these objectives.

On the other hand, the percentage of area for the view analysis is only sensitive to the stack options while the Glazing VLT and the WWR have no effect.

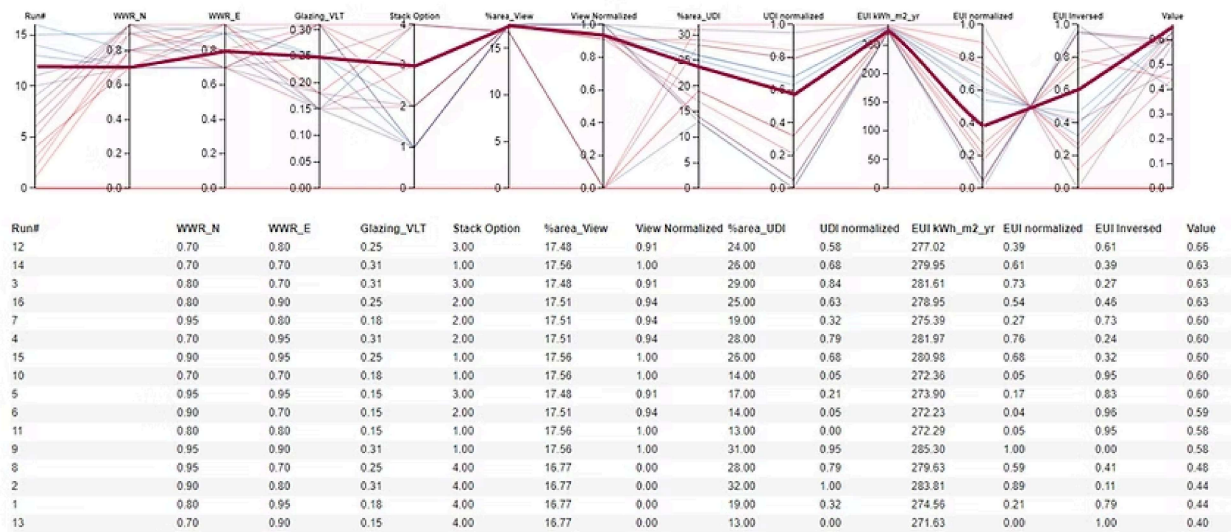
CONFLICTING OBJECTIVES



VALUE ASSESSMENT

Due to the conflicting objectives the data is then normalized/ given a rating from 0-1 such that their outputs become comparable. This data is then evaluated in order to recommend a best case scenario where the weightage /priority on UDI and EUI is 40% each and View is 20%. The below Parallel coordinates plot (PCP) is obtained by uploading the data into Design Explorer

Preferences: UDI = 40%, EUI= 40%, View=20%



RECOMMENDATION BASED ON OBJECTIVES- RUN#12

Best Option

Run#12

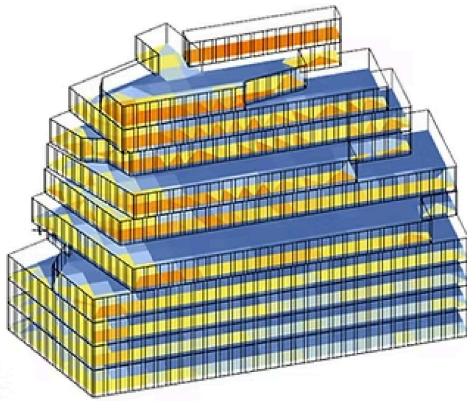
Inputs:

WWR_N= 0.7

WWR_E= 0.8

Glazing_VLT= 0.21

Stacking Option= 3



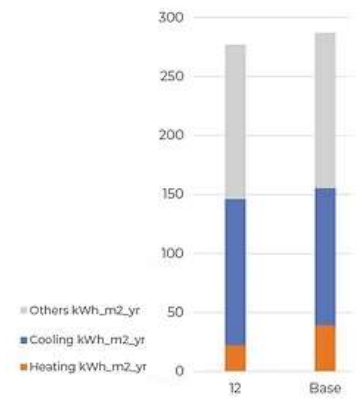
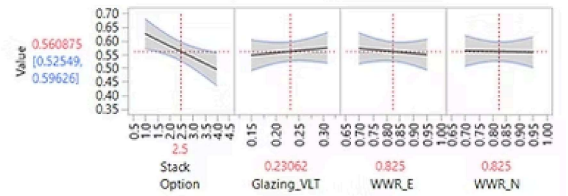
Parameter Effect

Column	Main Effect	Total Effect	.2	.4	.6	.8
Stack Option	0.914	0.934				
Glazing_VLT	0.024	0.039				
WWR_E	0.015	0.027				
WWR_N	0.001	0.002				

40% EUI, 40% UDI, 20% View

UDI= + 41.47%
EUI= - 3.51%

Input Sample



The value assessment therefore proves that having lower window to wall ratio on north and east sides and increasing the Glazing VLT to 0.21 while having the stack option to match the original building, helps in achieving the best case scenario within the design space.

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